

Vitamins

2025

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Water soluble vitamins

Vitamins **B complex and C**

- ❖ **Act as enzymatic cofactors**

- Folic acid and B12** act as coenzymes In hematopoiesis (formation of red blood cells) .
 - vitamin C** act as antioxidant.

- ❖ **The B vitamins are absorbed in** the intestine and transported In the portal circulation.

- The tissue stores** of most vitamins are **minimal**. The depletion occurs over several weeks in response to dietary restriction or increased the requirements as In pregnancy.

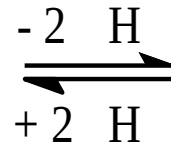
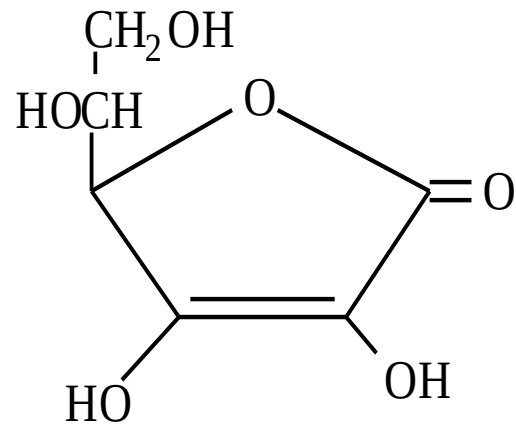
- ❖ Body stores of folic acid and vitamin B12 are more extensive than other B vitamins.

- ❖ **Toxic effects** are relatively uncommon, since excessive ingestion of water soluble vitamins is followed by rapid Loss of excess vitamins in the urine.

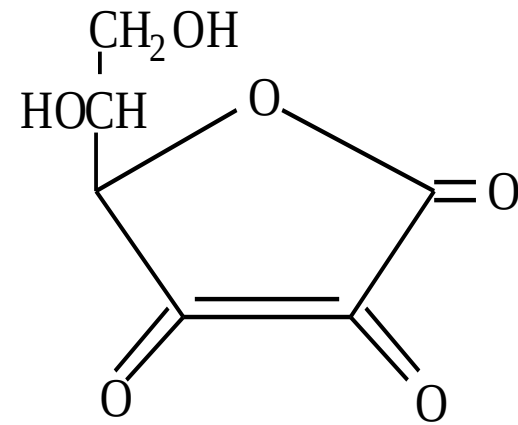
Vitamin C

- **Vitamin C** Is the **most labile vitamin** in food , it is easy to be destroyed.
- Much of its activity is lost through oxidation during preparation, cooking and storage.
- Vitamin C is **strong** reducing agent

Ascorbic acid



DehydroAscorbic Acid

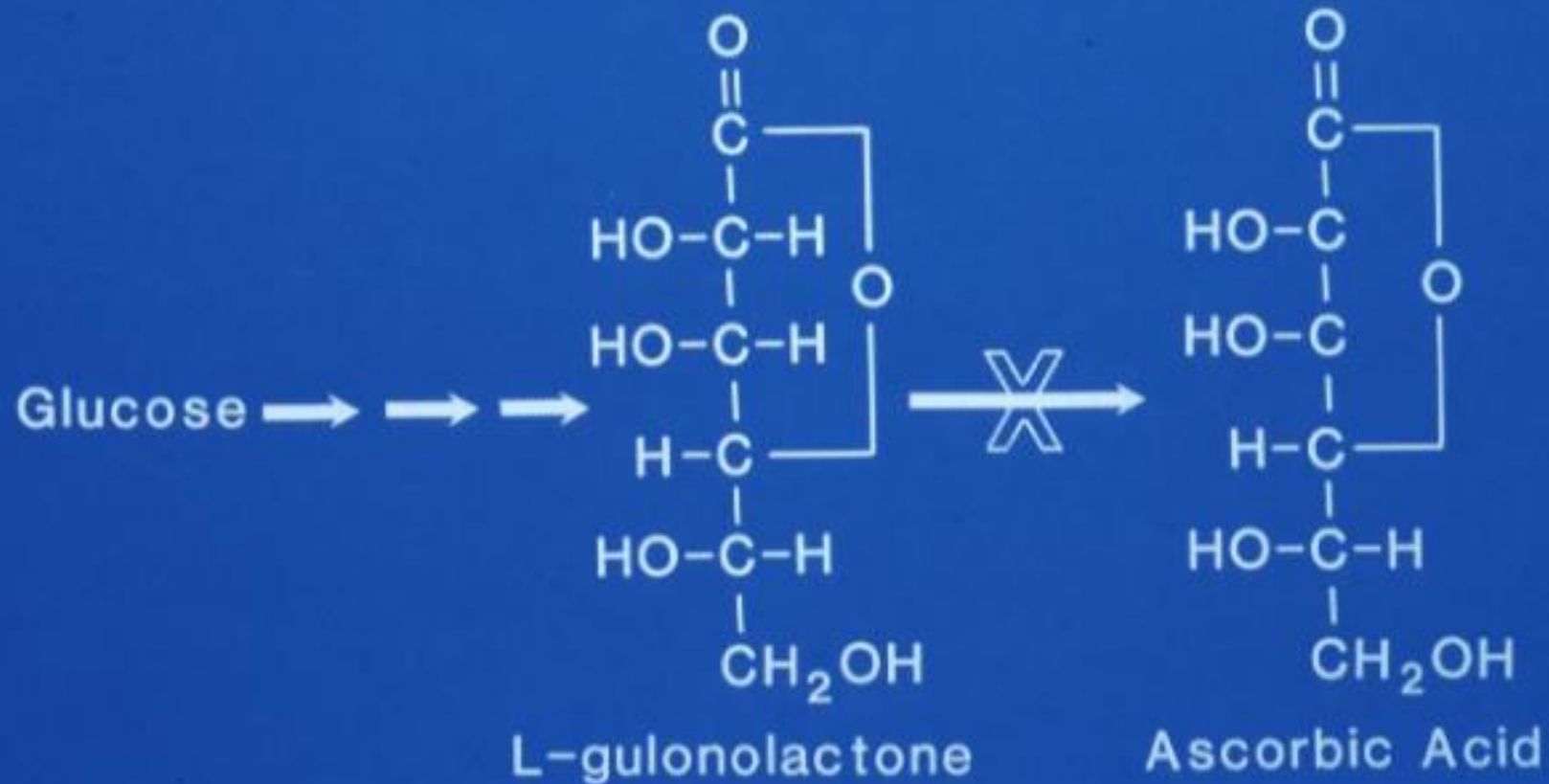


2 active form of vitamin C : ascorbic acid and dehydro-ascorbic acid

- In Animal tissues contains 90% L-ascorbic acid and 10% dehydro L-ascorbic acid. Both forms are active.
- Synthesized from glucose in plants and many animals (not in human) by L-gulonolactone oxidase
- Abundant in fruits especially citrus fruits (lemon, orange), and green leafy vegetables
- **RDA** 70 mg/day



ENZYME FOR ASCORBIC ACID SYNTHESIS IS MISSING IN MAN





VITAMIN C IS FOUND IN CITRUS FRUITS AND GREEN VEGETABLES



Vitamin C function

-Co enzyme of Hydroxylases enzyme

1-Formation of collagen protein:

- a) Ascorbic acid is essential for the conversion of the pro-collagen into collagen. Procollagen is a protein containing proline and lysine. Hydroxylation of both amino acids is catalyzed by **hydroxylase enzymes and by vitamin C as a coenzyme**. This converts procollagen into collagen.
- b) Collagen is essential for the synthesis of connective tissue, bone, cartilage and teeth..

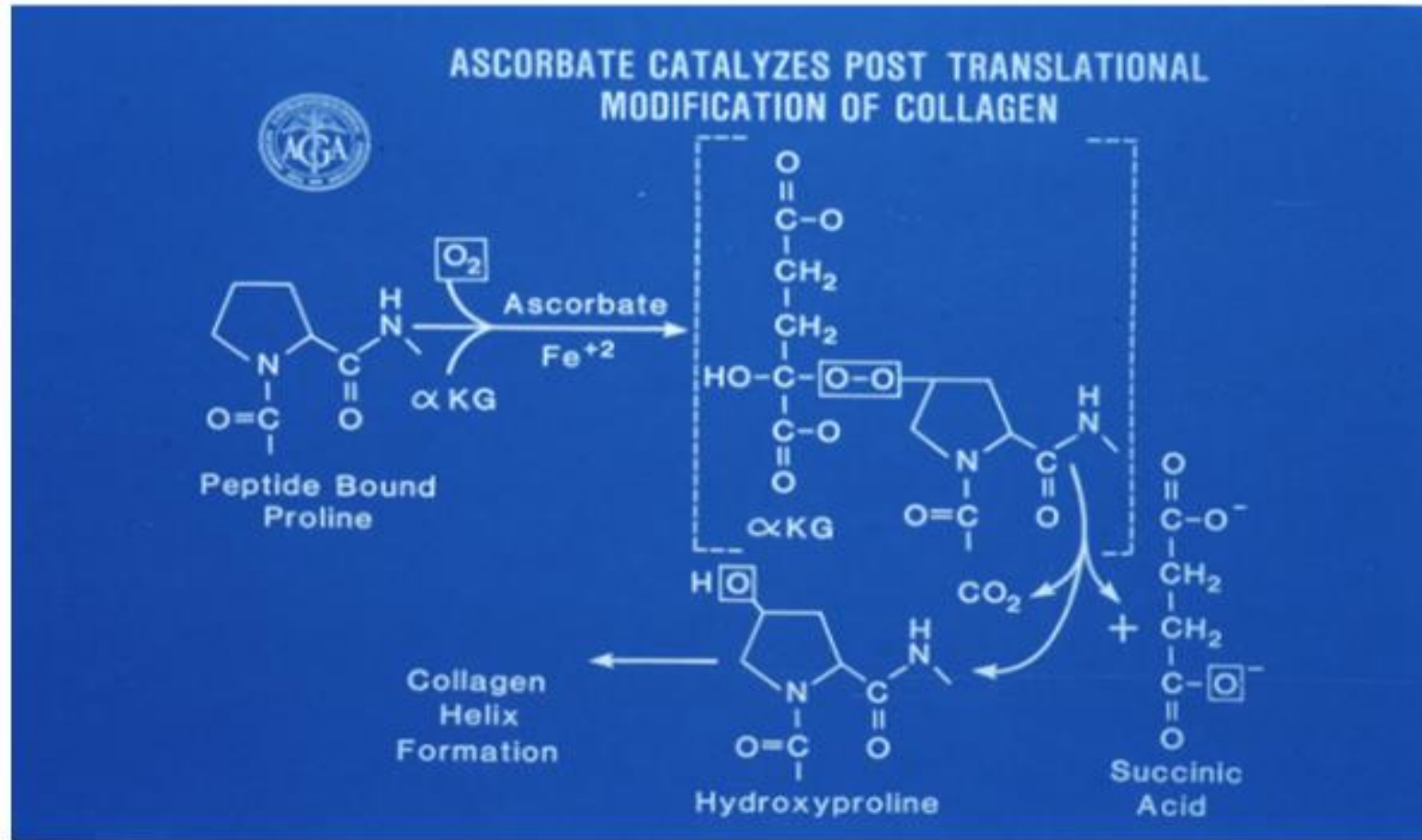
2-Ascorbic. acid also acts as a coenzyme for many **hydroxylase enzymes** In the pathways of:

- a) Bile acids synthesis.
- b) Osteocalcin synthesis.
- c) Carnitine synthesis: carnitine stimulates fatty acid oxidation In mitochondria((energy).
- d) Epinephrine synthesis.

3-Antioxidant : scavenge free radicals in aqueous media.

4-Enhance iron absorption: it reduce iron from ferric state to ferrous and keep it in ferrous state

collagen



Vitamin C functions

5-Detoxificate excess histamine by cleavage of imidazol ring (allergic response)

6- Detoxificate xenobiotics (heavy metals.....) .

- So it has a role to prevent or to share in treatment of cancer, cataract, atherosclerosis, hypertension, and common cold

Vitamin C Deficiency

If this store is depleted a disease called **Scurvy**, related to defective collagen synthesis

1-Manifestations due to decreased collagen formation:

- a) Bleeding into gum, muscles, joints, kidneys, gastrointestinal tract and pericardium.
- b) Defective formation of bone and teeth.
- c) Defective healing of wounds

2. Anemia: due to decreased absorption of iron and bleeding.

3. Manifestations due to decreased neurotransmitters (epinephrine and norepinephrine):

Behavioral changes. Severe emotional disturbances.

4. Manifestations due to decrease carnitine and fatty acids oxidation:

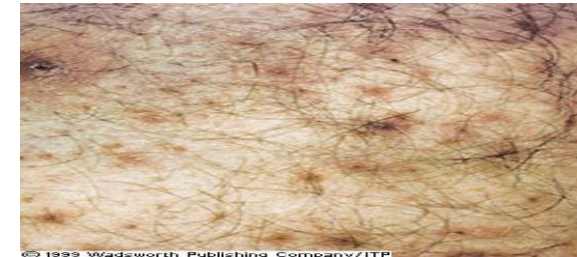
General weakness

Vitamin C Deficiency

Soft swollen gum



Pinpoint Bleeding from Scurvy



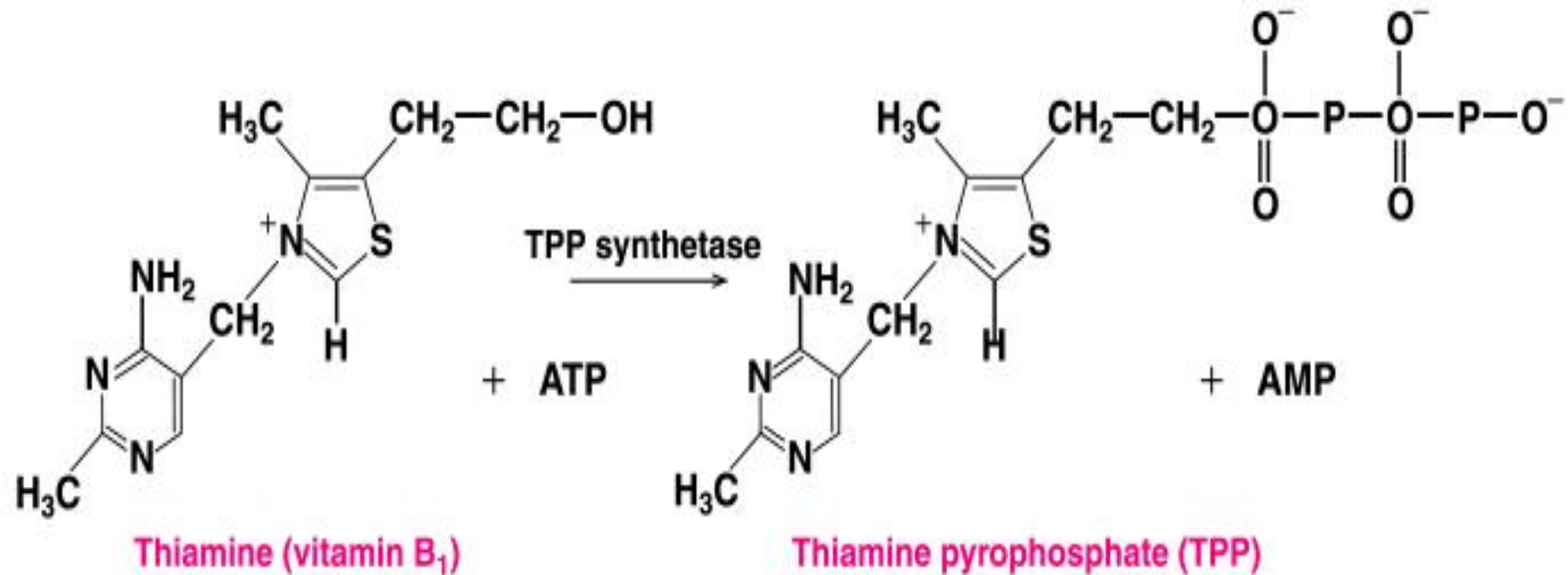
bruises



Vitamin C. Hypervitaminosis

- Increase iron absorption (**Siderosis**): Fe accumulate in tissues.
- forming O and OH free radicals , lead to oxidation of LDL (myocardial infarction), Free radicals convert ferrous to ferric
- **Increase production of oxalate** favoring deposition of Ca-oxalate stones in kidney and bladder

Vitamin B1 (Thiamin)



Vit B1 : is a Pyrimidine ring attached to thiazole ring by methylene bridge

Thiamin is rapidly converted to its active form, **thiamin pyrophosphate (TPP)**.

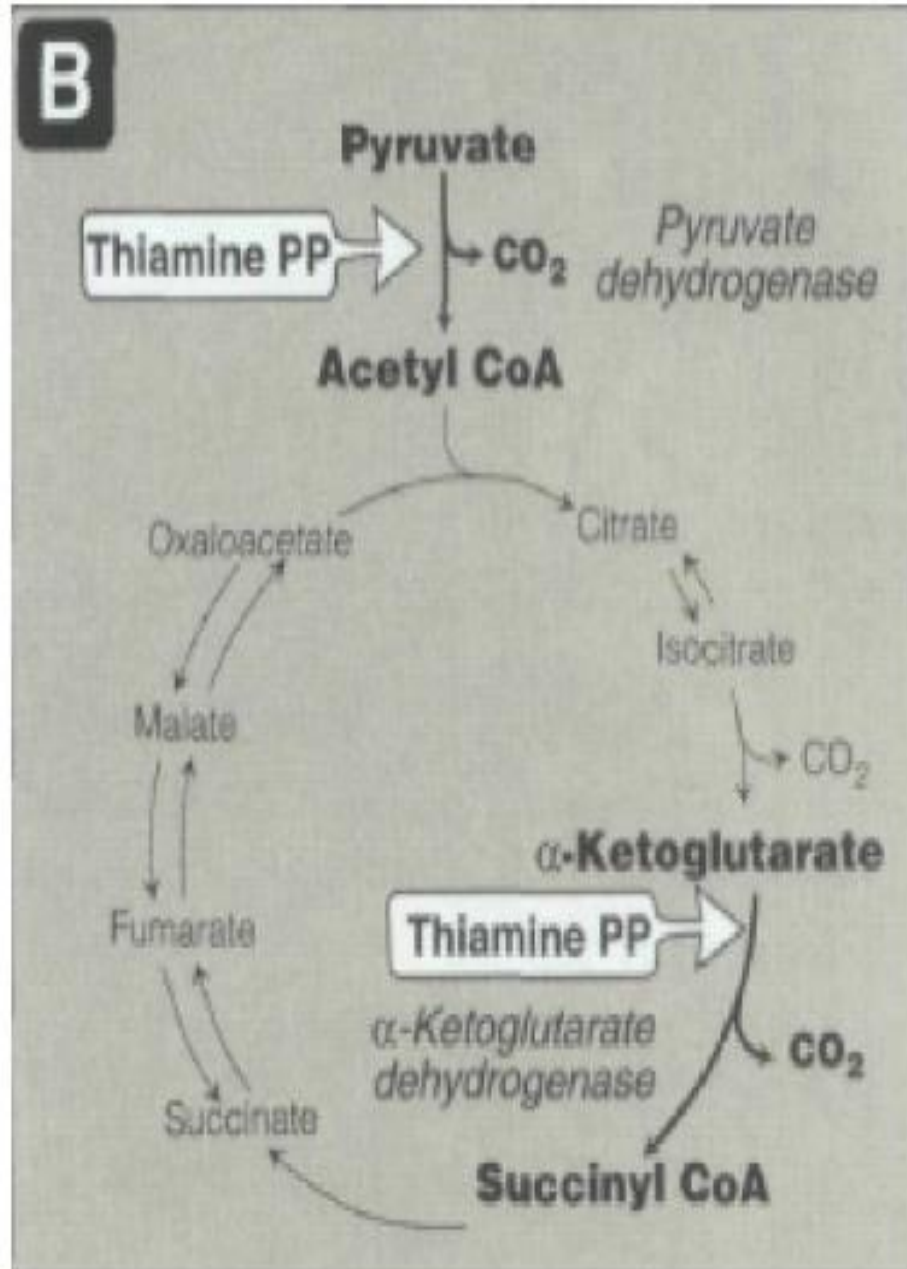
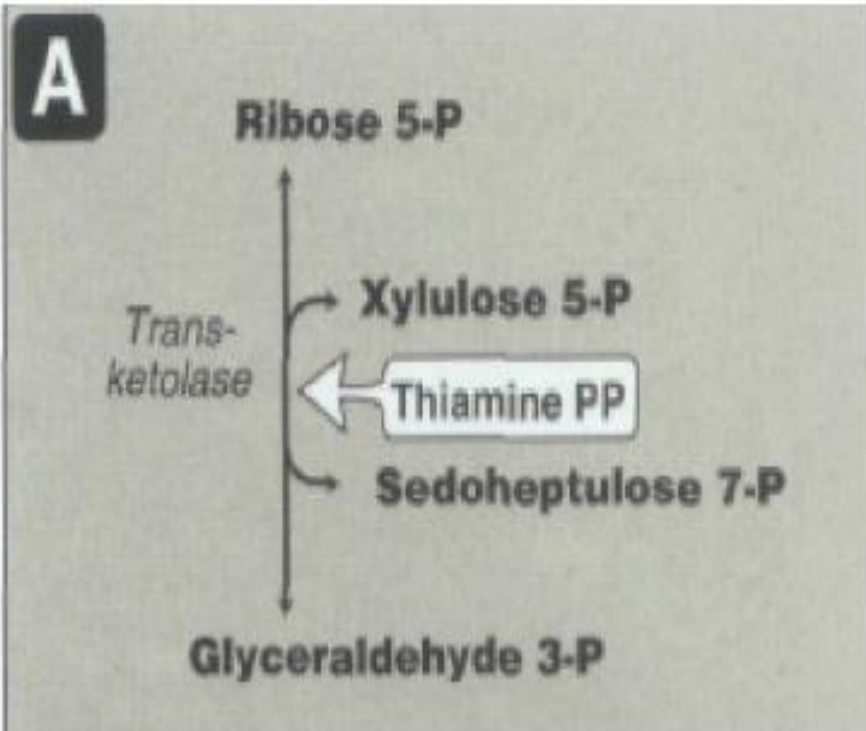
Vitamin B-1(Thiamin)

- Active form of B1** = Thiamin diphosphate ,also called thiamin pyrophosphate(**TPP**), .
- Formation of **TPP** needs **thiamin kinase** enzyme, which is present in liver, red cells and nervous tissue.
- It is destroyed by heat
- **Sources:** whole grains , watermelon , salmon , meat.
- **RDA** 1—1.5 mg

2-also in **branched-chain ketoacids dehydrogenase** involved in metabolism of leucine, isoleucine, and valine.

3- coenzyme for **transketolase**, in the pentose phosphate pathway (HMP)

4-It is essential for the process of nerve conduction



Thiamin deficiency

Carbohydrate with low thiamin (eg : polished rice) , highly refined foods (eg : sugar and white flour) , chronic alcoholism (alcohol inhibits thiamin absorption).

glucose metabolism is impaired and there is significant accumulation of pyruvic acid , lactic acid (Lethal) , pentose sugar , and ketoglutarate derivatives of branched amino acids.(BAA are transaminated with K.glutarate to generate glutamate)

Thiamin deficiency can result in distinct syndromes:

1- Beriberi :

symptom: peripheral neuropathy ,exhaustion , anorexia , edema, constipation , and cardiovascular , and muscular degeneration

2-Wernicke's encephalopathy with Korsakoff's psychosis, which is associated especially with alcohol.

Symptoms:

Mental confusion , Defective memory, ataxia, and loss of eye coordination, Depression.

Vitamin B-1

- **Toxicity**

vit B-1 is nontoxic even at high dosages

- **Estimation:**

Erythrocyte transketolase is used as a measure of thiamine deficiency

Vitamin B2 (Riboflavin)

-It is formed of **flavin ring** attached to **ribitol**.
Flavin ring is (a yellow fluorescent pigment)

-it is relatively heat stable

-photosensitive , i.e. decomposes with light.

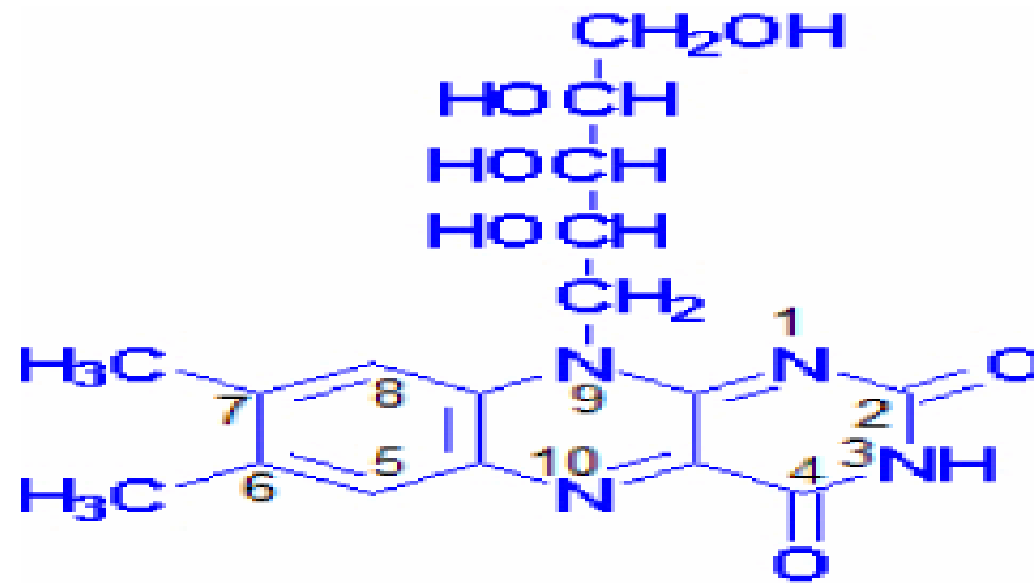
-The main dietary sources of riboflavin:

The main dietary sources of riboflavin are milk and dairy products . also present in green vegetables , almonds

- because of its intense yellow color, riboflavin is widely used as a food additive.

- **RDA** : 1.2—1.7 mg

Riboflavin (Vitamin B₂)



Riboflavin structure

Active forms of riboflavin:

Riboflavin enters in the structure of flavin mononucleotide (**FMN**) and flavin adenine dinucleotide (**FAD**), both are hydrogen carriers

-**FMN** is formed by phosphorylation of riboflavin by ATP (by intestinal flavin kinase enzyme).

-**FAD** is formed by the transfer of an AMP moiety from ATP to FMN

Function of Riboflavin

1 -**Both FMN and FAD are coenzyme** for flavoenzymes. They act as hydrogen (or electron) carriers in oxidation reduction reactions, these include the mitochondrial respiratory chain, key enzymes in fatty acid and amino acid oxidation, and the citric acid.

FMN.....FMNH₂

FADFADH₂ electron

flavoproteins contain one or more metal eg molybdenum and iron) so called **metalloflavoproteins** .

Vitamin B2 Deficiency

Causes of vit B2 deficiency :

1-phototherapy for treatment of hyper bilirubinemia in new born infants lead to riboflavin deficiencies

2- some drugs such as chlorpromazin, and alcohols decrease absorption

-

Riboflavin deficiency lead to :

1-**Ocular disturbances**: Photophobia .

2- **Cheilosis** (fissuring at the corners of the mouth).

3- **Glossitis** (inflammation of tongue)

4. **Dermatitis** (Inflammation of the skin).

-Estimation :

Erythrocyte glutathione reductase activity is used as an assay of vitamin B2 status

Riboflavin Deficiency: Ariboflavinosis

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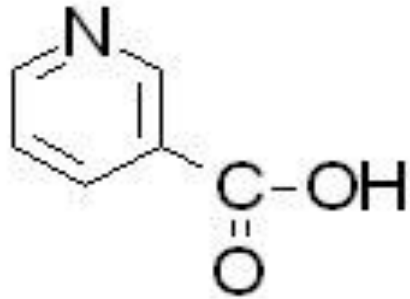
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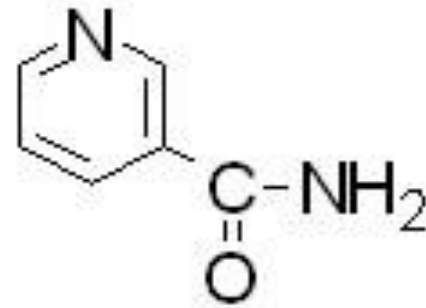
(b)

(a) A Colour Atlas and Text of Nutritional Disorders by Dr. Donald
D. McLaren/McGraw-Hill Europe Ltd.;
(b) Dr. P. Mancoske/Science Photo Library/Photo Researchers, Inc.

Vitamin B3 (Niacin)



Nicotinic acid



Nicotin amide

Two compounds nicotinic acid and nicotinamide, have the biological activity of niacin.

Sources:

- Synthesized in the body from the essential amino acid (**tryptophan**). This conversion requires vitamin B6 as a coenzyme.

Tryptophan (60mg) → niacin(1mg)

- Also present in Green leaf vegetables , liver , meat, coffee , tea.

RDA 16—34mg

Function of vitamin B3

1. Formation of (NAD) and (NADP) “active form of vitamin B3”: Niacin is essential for the formation of the **coenzymes** nicotinamide adenine dinucleotide(NAD) and nicotinamide adenine dinucleotide phosphate (NADP.).

These two coenzymes function as hydrogen carriers and they are essential for many oxidation-reduction reactions in carbohydrate, protein and lipid metabolism.

NAD⁺ (Oxidized form)

NADH (Reduced form)

NADP⁺ (Oxidized form)

NADPH (Reduced form)

2-NAD is a source of ADP-ribose which important for:

a) ADP-ribosylation of protein.

b) Poly ADP ribosylation of nucleoproteins Involved in **DNA repair mechanism.**

Function of vitamin B3

3- Lowering plasma cholesterol:

Niacin lowers plasma cholesterol concentration. This is due to inhibition of flow of free fatty acids (FFA) from adipose tissue

VitaminB3 Deficiency

Causes:

- 1) **Corn is deficient in both niacin and tryptophan.** Thus people who depend on corn as a major source of protein as some farmers develop pellagra.
- 2) **Hartnup's disease:** It is a hereditary disease in which there is defect in tryptophan absorption from intestine
- 3) **(malignant carcinoid syndrome):** In which large quantities of tryptophan is converted to serotonin. This occurs on the expense of niacin synthesis.
- 4) **Isoniazid:** It is a drug that is used in treatment of tuberculosis. Isoniazid binds with B6

VitaminB₃ Deficiency

- Lead to **Pellagra**:
4D (dementia , dermatitis , diarrhea , death)
gastrointestinal disturbance, loss of appetite
,headache, insomnia, mental depression , fatigue,
aches, and pains , nervousness, irritability



Vitamin B3 TOXICITY

Nicotinic acid has been used to treat hyperlipidemia when of the order of 1-6 g/d are required, causing dilatation of blood vessels and flushing, along with skin irritation. (not nicotine amide)

-also in case intakes of both nicotinic acid and nicotinamide in excess of 500 mg/d also cause liver damage.